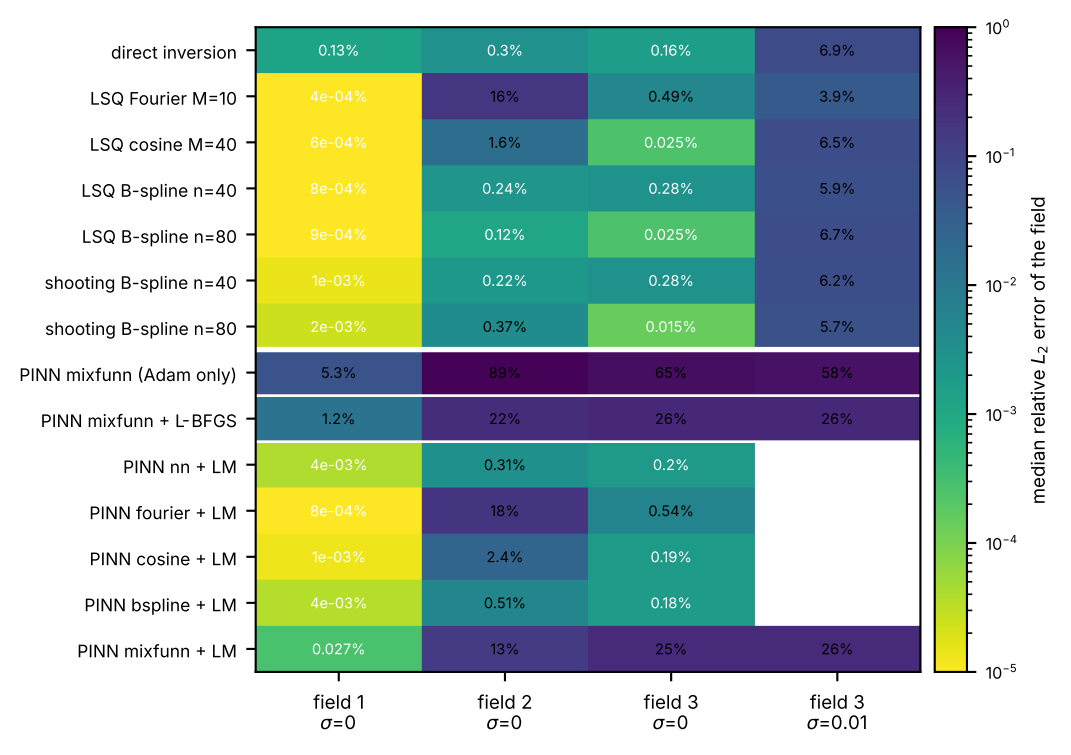
direct inversion
LSQ Fourier M=10
LSQ cosine M=40
LSQ B-spline n=40
LSQ B-spline n=80
shooting B-spline n=40
shooting B-spline n=80
PINN mixfunn (Adam only)
PINN mixfunn + L-BFGS
PINN nn + LM
PINN fourier + LM
PINN cosine + LM
PINN bspline + LM
PINN mixfunn + LM
0.13%
0.3%
0.16%
6.9%
4e-04%
16%
0.49%
3.9%
6e-04%
1.6%
0.025%
6.5%
8e-04%
0.24%
0.28%
5.9%
9e-04%
0.12%
0.025%
6.7%
1e-03%
0.22%
0.28%
6.2%
2e-03%
0.37%
0.015%
5.7%
5.3%
89%
65%
58%
1.2%
22%
26%
26%
4e-03%
0.31%
0.2%
8e-04%
18%
0.54%
1e-03%
2.4%
0.19%
4e-03%
0.51%
0.18%
0.027%
13%
25%
26%
field 1
σ=0
field 2
σ=0
field 3
σ=0
field 3
σ=0.01
median relative L_2 error of the field
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}